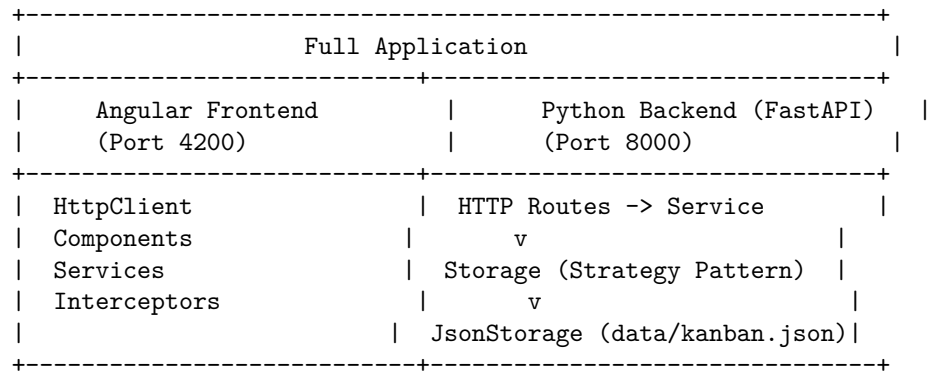


MVP Kanban Board Design Document

Intentions & Goals

Goal	Description
Build a minimalistic kanban board	Web-based task management with boards, columns, and tasks
Frontend in Angular	SPA using Angular framework
Backend in Python	REST API server using FastAPI
Flexible architecture	Strategy pattern for interchangeable storage/frontend
Testable	Mock frontend for backend testing without real Angular

Architecture



Strategy Pattern

Interface	Purpose	Implementations
StorageInterface	Database abstraction	JsonStorage, SQLiteStorage, PostgreStorage (future)
FrontendInterface	Frontend simulation	AngularMock (testing), AngularReal (production)

Key Design Decisions

Decision	Choice
Protocol	REST API with JSON (RFC 7807 errors)
Backend	FastAPI

Decision	Choice
Storage	JSON file (data/kanban.json)
ORM	None (direct JSON)
Auth	Skipped (development/testing)
IDs	Auto-integer
Data dir	Configurable via KANBAN_DATA_DIR env
Error on missing file	Fail with error

Service Layer Validation

Check	Behavior
Create task	column_id must exist
Create column	board_id must exist
Update task	target column_id must exist
Delete column	Cascade delete tasks in column
Delete board	Delete columns + tasks
Auto-cleanup	Remove invalid references

File Structure

```

backend/
+-- data/
|   +-- kanban.json
+-- src/
|   +-- kanban/
|       +-- __init__.py
|       +-- config.py
|       +-- models.py
|       +-- storage.py
|       +-- frontend.py
|       +-- service.py
|       +-- main.py
+-- tests/
|   +-- test_storage.py
+-- pyproject.toml

```

Sample Data

```

{
  "boards": [{"id": 1, "name": "My Board", "columns": [1, 2, 3]}],
  "columns": [
    {"id": 1, "name": "To Do", "position": 0},
    {"id": 2, "name": "In Progress", "position": 1},

```

```
    {"id": 3, "name": "Done", "position": 2}
  ],
  "tasks": [
    {"id": 1, "title": "Welcome Task", "description": "Drag me to move", "column_id": 1, "po
  ]
}
```